Example 5.23 Expand $\frac{1}{(3+2x)^2}$ in powers of x. Find a condition on x for which the expansion is valid.

Solution:
(Clearly we have to use the expansion of $(1+x)^{-2}$. So, we have to write $(3+2x)$ as $3\left(1+\frac{2x}{3}\right)$ and proceed.)

$$\begin{aligned}
\frac{1}{(3+2x)^2} &= \frac{1}{3^2\left(1+\frac{2x}{3}\right)^2} \\
&= \frac{1}{9}\left(1+\frac{2x}{3}\right)^{-2} \\
&= \frac{1}{9}(1+y)^{-2} \qquad \left(\text{where } \frac{2x}{3}=y\right) \\
&= \frac{1}{9}(1-2y+3y^2-4y^3+5y^4-\cdots), if |y|<1 \\
&= \frac{1}{9}\left(1-2\left(\frac{2x}{3}\right)+3\left(\frac{2x}{3}\right)^2-4\left(\frac{2x}{3}\right)^3+5\left(\frac{2x}{3}\right)^4-\cdots\right), |\frac{2x}{3}|<1 \\
&= \frac{1}{9}\left(1-\frac{4}{3}x+\frac{4}{3}x^2-\frac{32}{27}x^3+\frac{80}{81}x^4-\cdots\right) \\
\text{Thus, } \frac{1}{(3+2x)^2} &= \frac{1}{9}-\frac{4}{27}x+\frac{4}{27}x^2-\frac{32}{243}x^3+\frac{80}{729}x^4-\cdots, |x|<\frac{3}{2}
\end{aligned}$$

The expansion is valid if $|y|<1$. So, the expansion is valid if $|x|<\frac{3}{2}$.

We can find square root, cube root and other roots of any positive number by using binomial theorem. Let us see one such problem.

Example 5.24 Find $\sqrt[3]{65}$.

Solution:
We know that for $|x|<1$,

$$(1+x)^n = 1+nx+\frac{n(n-1)}{2!}x^2+\frac{n(n-1)(n-2)}{3!}x^3+\ldots$$

$$\begin{aligned}
\sqrt[3]{65} &= 65^{\frac{1}{3}} \\
&= (64+1)^{\frac{1}{3}} \\
&= 64^{\frac{1}{3}}\left(1+\frac{1}{64}\right)^{\frac{1}{3}} \\
&= 4\left(1+\frac{1}{64}\right)^{\frac{1}{3}}
\end{aligned}$$